

Dose-Specific Beta-Binomial Efficacy Model

Regular and IPDE efficacy with a dose-specific carryover parameter r

Purpose. This note describes the beta-binomial efficacy model with an IPDE carryover parameter. It is based on the current dose-specific JAGS implementation and identifies the specific change required to estimate r directly.

Implementation status. The current file `beta_binomial_dose_specific.jags` estimates independent regular and IPDE probabilities at doses 2 through J . It does not currently contain r . The model below is the r -based extension, replacing the independent IPDE probability prior at doses 2 through J .

1. Data and notation

For ordered dose levels $j = 1, \dots, J$, summarize observed efficacy outcomes as follows:

Quantity	Definition
$n_{R,j}; y_{R,j}$	Number treated and number with efficacy among regular patients at dose j .
$n_{I,j}; y_{I,j}$	Number treated and number with efficacy among IPDE patients at dose j .
$p_{R,j}$	Regular-patient efficacy probability at dose j .
$p_{I,j}$	IPDE efficacy probability at dose j .
r_j	Dose-specific IPDE efficacy carryover parameter at dose j .

Conditional on the two efficacy probabilities, the counts follow binomial sampling distributions:

$$y_{R,j} \mid p_{R,j} \text{ is } \text{Binomial}(n_{R,j}, p_{R,j})$$

$$y_{I,j} \mid p_{I,j} \text{ is } \text{Binomial}(n_{I,j}, p_{I,j})$$

2. Efficacy carryover model

Dose 1 has no preceding lower-dose reference. The regular and IPDE efficacy probabilities are therefore constrained to agree at dose 1:

$$p_{I,1} = p_{R,1}$$

For doses $j = 2, \dots, J$, define the IPDE efficacy probability by:

$$p_{I,j} = r_j + (1 - r_j) p_{R,j}$$

Equivalently, $1 - p_{I,j} = (1 - r_j)(1 - p_{R,j})$. Thus $r_j = 0$ gives no IPDE efficacy carryover; r_j close to 1 makes IPDE efficacy approach 1. This construction keeps $p_{I,j}$ in the unit interval whenever $p_{R,j}$ and r_j are in the unit interval.

3. Dose-specific Beta priors

Use distinct priors for baseline regular efficacy and for carryover:

$p_{R,j}$ is $\text{Beta}(a_P, j, b_P, j)$, $j = 1, \dots, J$
 r_j is $\text{Beta}(a_R, j, b_R, j)$, $j = 2, \dots, J$

For a $\text{Beta}(a, b)$ distribution, the prior mean is $a / (a + b)$ and the prior concentration is $a + b$. In the current R function, arguments named a_r and b_r are used for the Beta prior assigned to observed group probabilities. In the r extension, use separate a_P and b_P for baseline regular efficacy and reserve a_R and b_R for r.

4. Likelihood and posterior

Ignoring binomial coefficients, the efficacy likelihood is:

$$L(p_R, r) = \text{Product}_j [p_{R,j}^{y_{R,j}} (1-p_{R,j})^{(n_{R,j}-y_{R,j})}]$$
$$\text{times } [p_{I,j}^{y_{I,j}} (1-p_{I,j})^{(n_{I,j}-y_{I,j})}],$$

where $p_{I,1} = p_{R,1}$ and $p_{I,j} = r_j + (1-r_j)p_{R,j}$ for $j \geq 2$. The regular-only component is conjugate Beta-binomial. After IPDE observations are linked through r_j , the joint posterior of $p_{R,j}$ and r_j is not a closed-form Beta distribution, so JAGS samples it by MCMC.

5. JAGS specification for the r extension

```
model {
  for (j in 1:ndose) {
    p_regular[j] ~ dbeta(a_p[j], b_p[j])
    y_regular[j] ~ dbin(p_regular[j], n_regular[j])
  }
  p_ipde[1] <- p_regular[1]
  y_ipde[1] ~ dbin(p_ipde[1], n_ipde[1])
  for (j in 2:ndose) {
    r_eff[j] ~ dbeta(a_r[j], b_r[j])
    p_ipde[j] <- r_eff[j] + (1-r_eff[j]) * p_regular[j]
    y_ipde[j] ~ dbin(p_ipde[j], n_ipde[j])
  }
}
```

Monitor p_{regular} , p_{ipde} , and r_{eff} . The regular-patient efficacy curve is summarized by the posterior distribution of $p_{\text{regular},j}$.

6. Relation to the current JAGS code

Current dose-specific JAGS file	r-carryover extension
Independent Beta priors for $p_{\text{regular},j}$ and $p_{\text{ipde},j}$ at $j \geq 2$.	A Beta prior for $p_{\text{regular},j}$ and a Beta prior for r_j ; $p_{\text{ipde},j}$ is deterministic.
Allows the two groups to differ but does not parameterize the difference.	Separates baseline regular efficacy from the IPDE carryover effect.
Uses $a_{r,j}$ and $b_{r,j}$ for both observed group probabilities.	Uses $a_{P,j}$ and $b_{P,j}$ for baseline efficacy; $a_{R,j}$ and $b_{R,j}$ for r_j .

7. Interpretation and practical checks

Identifiability. Estimating r_j requires regular and IPDE observations at the same dose. With few IPDE patients, posterior intervals for r_j will be wide and prior-sensitive.

Dose 1. Do not estimate a separate r_1 : the model constrains $p_{I,1} = p_{R,1}$.

Alpha-generated data. If IPDE efficacy is generated as $p_{R,j} + \alpha p_{R,j-1}$, its equivalent carryover value is $r_j = (p_{I,j} - p_{R,j}) / (1 - p_{R,j})$, provided $p_{I,j}$ is below 1.

Posterior reporting. Report posterior mean or median, 95 percent credible interval, and the probability that $p_{R,j}$ exceeds a clinically chosen efficacy target.

Prior sensitivity. Compare the specified Beta priors by examining their effect on $p_{R,j}$, $p_{I,j}$, and r_j when IPDE sample sizes are small.

Summary. The r-based model makes the structural assumption that IPDE exposure can only preserve or increase efficacy relative to regular treatment at the same dose. If IPDE efficacy can be lower, use separate unconstrained $p_{R,j}$ and $p_{I,j}$ probabilities instead.